A Fourier–compactness argument for Erdős Problem 42

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Abstract

A proof of Erdős Problem 42 is presented. The argument reduces the problem to a one-sided Fourier lemma for Cayley graphs over prime cyclic finite fields. The main technical point is a two-colour weighted degree-one compactness/counting proposition. Since the corresponding results in Szegedy's paper are not stated in precisely this multicolour weighted form, the required bridge is proved from the exact ingredients used there: the ultraproduct Fourier factor, the degree-one generalized von Neumann estimate, the structure of functions measurable with respect to the Fourier sigma-algebra, and the dual embedding of degree-one limits. The remaining steps are the spectral passage from finite one-sided Fourier lower bounds to Fourier positivity on the compact limit, the construction of a continuous positive-definite representative, and a weighted positivity argument on connected compact abelian groups.

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1 Introduction and statement of the theorem

A finite set $S \subset \mathbb{Z}$ is called a Sidon set if all pairwise sums

$$s + s', \qquad s, s' \in S, \quad s \le s',$$

are distinct. Equivalently, every nonzero integer has at most one representation as an ordered difference $s - s'$ with $s, s' \in S$.

The problem considered here is the following question of Erdős, from [1]. Bloom's Erdős Problems entry is also cited for the current problem statement and discussion record [2].

Theorem 1.1 (Erdős Problem 42)**.** *For every $M \ge 1$ there is $N_0(M)$ such that, whenever $N \ge N_0(M)$ and $A \subset \{1, \dots, N\}$ is a Sidon set, there is a Sidon set $B \subset \{1, \dots, N\}$ with*

$$|B| = M$$

and

$$(A - A) \cap (B - B) = \{0\}.$$

Provenance and relation to the public discussion

The proof below is written as a self-contained argument for every fixed M. The public discussion attached to the Erdős Problems page contains relevant partial progress, reductions, and comments on the proposed solution, and those contributions should be separated from the new part of the present paper. The problem page records the cases $M = 1, 2, 3$ as partial progress in the comments and lists Zach Hunter and Daniil Sedov under "Additional thanks" [2]; these partial results are not used below. In the discussion thread, Harjas posted the proposed solution that motivated the present write-up [4]; qrdl formulated a special case of the one-sided Fourier key lemma and the resulting reduction to a random choice of B [5]; Thomas F. Bloom pointed out the Cauchy–Schwarz complexity-one structure of the forms $x_i - x_j$ [6]; Will Sawin described the regularity/compact-limit interpretation of the key lemma [7]; and Terence Tao gave the continuous compact positive-definite analogue and explained how it fits the compactness route [8]. Tao also noted the greedy reduction that removes the need to find a Sidon set directly [9]; Appendix A gives the elementary written proof of the extraction used here. The main contribution of this paper is the general fixed-M argument via Lemma 1.2, and especially the fully stated multicolour compactness/counting bridge in Proposition 4.5. The full discussion thread is cited as [3]; no unproved assertion from that thread is used as a black box.

Finite cyclic groups are written additively. For a prime p, the normalized Fourier transform on $\mathbb{F}_p$ is

$$\widehat{f}(\gamma) = \frac{1}{p} \sum_{x \in \mathbb{F}_p} f(x) e^{-2\pi i \gamma x / p}.$$

All expectations on finite groups are normalized. On compact abelian groups the convention is

$$\widehat{f}(\chi) = \int_G f(x) \overline{\chi(x)} \, dm_G(x).$$

The central auxiliary result is the following one-sided Cayley–Fourier lemma.

Lemma 1.2 (One-sided Cayley–Fourier lemma)**.** *Fix $K \ge 1$, $0 < \alpha < 1$, and $0 < \theta < 1$. There are $\varepsilon > 0$ and p_0 such that the following holds for every prime $p \ge p_0$.*

Let $F \subset \mathbb{F}_p$ satisfy

$$0 \in F, \qquad F = -F, \qquad |F| \le (1-\alpha)p,$$

and

$$\widehat{\mathbf{1}_F}(\gamma) \ge -\varepsilon \qquad (\gamma \ne 0).$$

Set

$$I_p = [1, \lfloor \theta p \rfloor] \subset \mathbb{F}_p.$$

Then

$$\mathbb{E}_{x_1,\dots,x_K \in I_p} \prod_{1 \le i < j \le K} \big(1 - \mathbf{1}_F(x_i - x_j)\big) > 0.$$

Equivalently, there exist distinct $x_1, \dots, x_K \in I_p$ such that

$$x_i - x_j \notin F \qquad (i \ne j).$$

2 Reduction to the one-sided Cayley–Fourier lemma

The first step is the reduction of Theorem 1.1 to Lemma 1.2.

Lemma 2.1 (Fourier lower bound for Sidon difference sets)**.** *Let $A \subset \{1, \dots, N\}$ be Sidon. Let $p > 2N$ be prime and view A as a subset of $\mathbb{F}_p$. Put*

$$F = A - A \subset \mathbb{F}_p.$$

Then

$$0 \in F, \qquad F = -F,$$

$$r_{A-A} = |A|\mathbf{1}_{\{0\}} + \mathbf{1}_{F \setminus \{0\}},$$

$$|F| = |A|(|A| - 1) + 1,$$

and, for every $\gamma \in \widehat{\mathbb{F}_p}$,

$$\widehat{\mathbf{1}_F}(\gamma) \ge -\frac{|A| - 1}{p}.$$

Moreover

$$|A| \le 1 + \sqrt{2N} \qquad \text{and} \qquad |F| \le 2N - 1.$$

Proof. Since $A \subset [1, N]$, all integer differences $a - a'$ lie in $(-(N-1), N-1)$. If two such differences are congruent modulo p and $p > 2N$, then their difference is a multiple of p of absolute value strictly less than p, hence is zero. Thus equality of differences modulo p is exactly equality of differences in $\mathbb{Z}$.

Let

$$r_{A-A}(x) = |\{(a, a') \in A^2 : a - a' = x\}|.$$

The diagonal pairs give $r_{A-A}(0) = |A|$. Since A is Sidon, every nonzero difference has at most one ordered representation. Hence

$$r_{A-A} = |A|\mathbf{1}_{\{0\}} + \mathbf{1}_{F \setminus \{0\}}.$$

Equivalently,

$$\mathbf{1}_F = r_{A-A} - (|A| - 1)\mathbf{1}_{\{0\}}.$$

For the normalized Fourier transform,

$$\widehat{r_{A-A}}(\gamma) = p\,|\widehat{\mathbf{1}_A}(\gamma)|^2,$$

and

$$\widehat{\mathbf{1}_{\{0\}}}(\gamma) = \frac{1}{p}.$$

Therefore

$$\widehat{\mathbf{1}_F}(\gamma) = p|\widehat{\mathbf{1}_A}(\gamma)|^2 - \frac{|A|-1}{p} \geq -\frac{|A|-1}{p}.$$

The same difference representation shows that the $|A|(|A|-1)$ ordered nonzero differences are all distinct, and then

$$|F| = |A|(|A|-1)+1.$$

Equivalently, considering only positive differences gives

$$\binom{|A|}{2} \leq N-1.$$

Thus $|A| \leq 1+\sqrt{2N}$, and

$$|F| = |A|(|A|-1)+1 \leq 2N-1.$$

□

Proof of Theorem 1.1 assuming Lemma 1.2. Let $R(M)$ be an integer such that every set of at least $R(M)$ integers contains a Sidon subset of size M. The existence of $R(M)$ is proved in Appendix A.

Apply Lemma 1.2 with

$$K = R(M), \qquad \alpha = \frac{1}{2}, \qquad \theta = \frac{1}{8}.$$

Let $\varepsilon > 0$ and p_0 be the corresponding constants.

Let N be sufficiently large in terms of M, and let $A \subset [1,N]$ be Sidon. By Bertrand's postulate, applied to $4N$, there is a prime

$$4N \leq p \leq 8N$$

for all $N \geq 1$ after harmless adjustment of the endpoint convention. In particular $p > 2N$. View A as a subset of $\mathbb{F}_p$, and set

$$F = A - A \subset \mathbb{F}_p.$$

By Lemma 2.1,

$$|F| \leq 2N-1 < 2N \leq \frac{p}{2}.$$

Also

$$\widehat{\mathbf{1}_F}(\gamma) \geq -\frac{|A|-1}{p} \geq -\frac{\sqrt{2N}}{4N}.$$

For N sufficiently large this is at least $-\varepsilon$.

Lemma 1.2 yields distinct elements

$$x_1, \ldots, x_K \in [1, \lfloor p/8 \rfloor] \subset [1,N]$$

such that

$$x_i - x_j \notin F \qquad (i \neq j)$$

in $\mathbb{F}_p$. Let

$$X = \{x_1, \dots, x_K\}.$$

By the choice of $K = R(M)$, X contains a Sidon subset B of size M.

It remains to pass from $\mathbb{F}_p$ back to the interval. If

$$b - b' = a - a'$$

in $\mathbb{Z}$, with $b, b' \in B$ and $a, a' \in A$, then the same equality holds modulo p. Since $B \subset X$, the construction gives

$$b - b' \notin A - A$$

modulo p unless $b = b'$. Hence $b = b'$, and then $a - a' = 0$, so $a = a'$. Therefore

$$(A - A) \cap (B - B) = \{0\}.$$

□

3 Proof of the one-sided Cayley–Fourier lemma by contradiction

Assume Lemma 1.2 fails for some fixed

$$K \geq 1, \qquad 0 < \alpha < 1, \qquad 0 < \theta < 1.$$

Then there are primes $p_n \to \infty$, numbers $\varepsilon_n \downarrow 0$, and sets $F_n \subset \mathbb{F}_{p_n}$ such that

$$0 \in F_n, \qquad F_n = -F_n, \qquad |F_n| \leq (1-\alpha)p_n,$$

$$\widehat{\mathbf{1}_{F_n}}(\gamma) \geq -\varepsilon_n \qquad (\gamma \neq 0),$$

but, with

$$I_n = [1, \lfloor \theta p_n \rfloor] \subset \mathbb{F}_{p_n},$$

one has

$$\mathbb{E}_{x_1,\dots,x_K \in I_n} \prod_{i<j} \big(1 - \mathbf{1}_{F_n}(x_i - x_j)\big) = 0$$

for every n.

Set

$$u_n = \mathbf{1}_{I_n}, \qquad h_n = \mathbf{1}_{F_n}.$$

The multicolour compactness/counting proposition proved in Section 4 gives a connected compact abelian group G, functions $u, h : G \to [0,1]$, and

$$\int_G u \, dm_G = \theta, \qquad \int_G h \, dm_G \leq 1 - \alpha.$$

The finite Fourier lower bound passes to the limit and makes h Fourier-positive. By the analytic lemmas in Section 6, the function h may be replaced by a continuous positive-definite representative. Then, for

$$d\nu = \frac{u \, dm_G}{\theta},$$

the coloured counting statement gives

$$\lim_{n\to\infty} \mathbb{E}_{x_1,\dots,x_K\in I_n} \prod_{i<j}\big(1-h_n(x_i-x_j)\big)$$
$$= \int_{G^K} \prod_{i<j}\big(1-h(y_i-y_j)\big)\, d\nu^K(\mathbf{y}).$$

The weighted positivity lemma in Section 7 shows that the last integral is strictly positive, contradicting the vanishing of the finite averages. Thus Lemma 1.2 holds.

The remainder of the paper supplies the details used in this contradiction argument.

4 The two-colour degree-one compactness/counting proposition

This section is deliberately explicit about what is, and is not, taken from Szegedy's paper [13]. In Section 4, formula (1), Szegedy defines the coloured density

$$t(L,F) = \mathbb{E}_{\mathbf{x}\in A^d} \prod_{\ell=1}^{t} f_\ell(L_\ell(\mathbf{x})),$$

where $F = (f_1,\dots,f_t)$ is a tuple of bounded functions. Definition 4.1 defines true complexity through these coloured densities. However, Szegedy's Theorem 4 states continuity for the monochromatic map $f \mapsto t(L,f)$, and Proposition 6.2 is also stated for $t(L,f)$. Those two statements therefore do not, as written, directly give the weighted two-colour proposition needed here. The proof below supplies the missing bridge.

The exact Szegedy ingredients used are the following. Proposition 6.1 identifies linear characters of an ultraproduct as ultralimits of finite linear characters. Lemma 6.3 identifies the Fourier sigma-algebra $\mathcal{F}(A)$ as the characteristic factor for the U^2-seminorm: $\|g\|_{U^2} = 0$ if and only if g is orthogonal to $L^2(\mathcal{F}(A))$. Theorem 5 represents every bounded $\mathcal{F}(A)$-measurable function on a second-countable compact abelian factor. Corollary 7.2 gives the dual embedding for degree-one limits; in the present prime-order situation this forces the limiting compact group to be connected. Proposition 6.2 is not used as a black box for the coloured result; its proof motivates the factor reduction, which is spelled out below.

Definition 4.1 (Cauchy–Schwarz complexity at most one)**.** *A finite system of integer linear forms*

$$L = (L_1,\dots,L_t)$$

in variables $x_1,\dots,x_d$ has Cauchy–Schwarz complexity at most 1 if, for every j, the remaining forms can be partitioned into two classes

$$\{L_i : i \neq j\} = \mathcal{C}_1 \sqcup \mathcal{C}_2$$

such that

$$L_j \notin \operatorname{span}_{\mathbb{Q}}(\mathcal{C}_1) \quad \textit{and} \quad L_j \notin \operatorname{span}_{\mathbb{Q}}(\mathcal{C}_2).$$

Lemma 4.2 (Generalized von Neumann estimate, complexity one)**.** *Let $L = (L_1,\dots,L_t)$ be a fixed system of integer linear forms of Cauchy–Schwarz complexity at most 1. Let $A_n = \mathbb{F}_{p_n}$ with $p_n \to \infty$. Then, after discarding finitely many n, for every j and every choice of functions $f_{\ell,n} : A_n \to \mathbb{C}$ with $|f_{\ell,n}| \le 1$,*

$$\left| \mathbb{E}_{\mathbf{x}\in A_n^d} \prod_{\ell=1}^{t} f_{\ell,n}(L_\ell(\mathbf{x})) \right| \le \|f_{j,n}\|_{U^2(A_n)}.$$

Consequently, if $A = \prod_\omega A_n$ is a Szegedy ultraproduct and $f_\ell \in L^\infty(A)$ with $|f_\ell| \le 1$, then

$$\left| \int_{A^d} \prod_{\ell=1}^{t} f_\ell(L_\ell(\mathbf{x}))\, d\mathbf{x} \right| \le \|f_j\|_{U^2(A)}.$$

Proof. For finite vector spaces this is the $s = 1$ case of the generalized von Neumann theorem for systems of finite Cauchy–Schwarz complexity; see Green–Tao [11, Proposition 7.1]. The only harmless point in the present formulation is that the forms have integer coefficients. For a fixed system L, the rational non-containment conditions defining Cauchy–Schwarz complexity remain valid over $\mathbb{F}_{p_n}$ for all sufficiently large primes p_n, because only finitely many integer determinants are involved.

The quoted proposition is proved by applying Cauchy–Schwarz once for each of the two classes in the defining partition for L_j. In this complexity-one case the resulting expression is exactly bounded by the U^2-norm of $f_{j,n}$; all other functions remain bounded by 1. Thus the displayed finite estimate follows.

For the ultraproduct statement, represent the bounded measurable f_ℓ up to null sets as ultralimits of bounded functions $f_{\ell,n}$ on A_n, using the ultralimit measurability statement in Szegedy's Section 5. Apply the finite estimate and take the ultralimit. The ultraproduct U^2-seminorm is, by definition, the ultralimit of the finite U^2-norms for internal representatives and then extends by approximation to bounded measurable functions. □

Lemma 4.3 (Fourier-factor reduction for two colours)**.** *Let $A = \prod_\omega \mathbb{F}_{p_n}$ be a Szegedy ultraproduct with $p_n \to \infty$, and let $U, H \in L^\infty(A)$ satisfy $0 \le U, H \le 1$. Let*

$$U_0 = \mathbb{E}(U \mid \mathcal{F}(A)), \qquad H_0 = \mathbb{E}(H \mid \mathcal{F}(A)).$$

For every Cauchy–Schwarz complexity-one system $L = (L_1, \dots, L_t)$ and every colour assignment $C_\ell \in \{U, H\}$, with $C_{\ell,0} = U_0$ or H_0 according to the same colour, one has

$$\int_{A^d} \prod_{\ell=1}^{t} C_\ell(L_\ell(\mathbf{x}))\, d\mathbf{x} = \int_{A^d} \prod_{\ell=1}^{t} C_{\ell,0}(L_\ell(\mathbf{x}))\, d\mathbf{x}.$$

Proof. Write

$$C_\ell = C_{\ell,0} + R_\ell.$$

By Szegedy's Lemma 6.3, each residual R_ℓ is orthogonal to $L^2(\mathcal{F}(A))$. Hence

$$\|R_\ell\|_{U^2(A)} = 0.$$

Expanding the left-hand side multilinearly, the term with no residuals is exactly the right-hand side. Every other term contains at least one factor $R_j(L_j(\mathbf{x}))$. All factors are bounded by 1, and Lemma 4.2 gives absolute value at most

$$\|R_j\|_{U^2(A)} = 0.$$

Thus all residual terms vanish. □

Lemma 4.4 (Simultaneous representation on one compact factor)**.** *Let $A = \prod_\omega \mathbb{F}_{p_n}$ be as above, and let*

$$U_0, H_0 \in L^\infty(\mathcal{F}(A)), \qquad 0 \le U_0, H_0 \le 1.$$

Then there are a second-countable compact abelian group G, a Szegedy-continuous surjective measure-preserving homomorphism

$$\pi : A \to G,$$

and Borel functions $u, h : G \to [0,1]$ such that

$$U_0 = u \circ \pi, \qquad H_0 = h \circ \pi$$

almost everywhere. Moreover G may be chosen connected.

Proof. Apply Szegedy's Theorem 5 to the single bounded $\mathcal{F}(A)$-measurable function

$$W_0 = U_0 + iH_0.$$

It gives a second-countable compact abelian group G, a Szegedy-continuous surjective measure-preserving homomorphism $\pi : A \to G$, and $w \in L^\infty(G)$ such that

$$W_0 = w \circ \pi$$

almost everywhere. Set $u = \operatorname{Re} w$ and $h = \operatorname{Im} w$, modifying on a Haar-null set so that $0 \le u, h \le 1$ everywhere.

It remains to record connectedness. In the proof of Szegedy's Theorem 5, G is generated by the countable family of characters occurring in the Fourier expansion of W_0, and the dual map

$$\widehat{\pi} : \widehat{G} \hookrightarrow \widehat{A}$$

is injective. By Szegedy's Proposition 6.1,

$$\widehat{A} = \prod_\omega \widehat{\mathbb{F}_{p_n}} \cong \prod_\omega \mathbb{Z}/p_n\mathbb{Z}.$$

This ultraproduct has no nonzero torsion. Indeed, if $q\gamma = 0$ and $\gamma = (\gamma_n)_\omega$, then $q\gamma_n = 0$ for ω-almost all n; since $p_n \to \infty$, q is invertible modulo p_n for ω-almost all n, so $\gamma_n = 0$ for ω-almost all n. Hence $\gamma = 0$. Thus $\widehat{G}$ is torsion-free. By Pontryagin duality, a compact abelian group is connected if and only if its dual group is torsion-free, so G is connected. □

Proposition 4.5 (Multicolour degree-one compactness/counting)**.** *Let $p_n \to \infty$ be primes, let $A_n = \mathbb{F}_{p_n}$, and let*

$$u_n, h_n : A_n \to [0,1].$$

After passing to a subsequence and fixing a non-principal ultrafilter ω, there exist a single connected second-countable compact abelian group G, measurable functions

$$u, h : G \to [0,1],$$

and a factor map

$$\pi : \prod_\omega A_n \to G$$

with the following properties.

(a) For every Cauchy–Schwarz complexity-one system $L = (L_1, \dots, L_t)$ of integer linear forms and every colour assignment

$$c_{\ell,n} \in \{u_n, h_n\},$$

with corresponding limit colour $c_\ell \in \{u, h\}$, the finite coloured densities converge and satisfy

$$\lim_{n\to\infty} \mathbb{E}_{\mathbf{x}\in A_n^d} \prod_{\ell=1}^t c_{\ell,n}(L_\ell(\mathbf{x})) = \int_{G^d} \prod_{\ell=1}^t c_\ell(L_\ell(\mathbf{y}))\, dm_G^d(\mathbf{y}).$$

(b) *The same compact group G and the same factor map π represent both colours: if*

$$U = \lim_{\omega} u_n, \qquad H = \lim_{\omega} h_n,$$

then

$$\mathbb{E}(U \mid \mathcal{F}) = u \circ \pi, \qquad \mathbb{E}(H \mid \mathcal{F}) = h \circ \pi$$

almost everywhere on the ultraproduct, where $\mathcal{F}$ is Szegedy's Fourier sigma-algebra.

(c) *The group G is connected; equivalently, $\widehat{G}$ is torsion-free.*

(d) *For every $\chi \in \widehat{G}$, there is a representing sequence of characters $\gamma_n \in \widehat{A_n}$ such that*

$$\widehat{h}(\chi) = \lim_{\omega} \widehat{h_n}(\gamma_n).$$

The analogous identity holds for u_n, u.

Proof. First pass to a subsequence along which every coloured density associated with a Cauchy–Schwarz complexity-one integer system and colours from $\{u_n, h_n\}$ converges. This diagonal selection is possible because there are only countably many finite integer systems and finite colour assignments. It also makes the one-form densities $\mathbb{E}u_n$ and $\mathbb{E}h_n$ converge.

Let

$$A = \prod_{\omega} A_n, \qquad U = \lim_{\omega} u_n, \qquad H = \lim_{\omega} h_n.$$

Let $\mathcal{F} = \mathcal{F}(A)$ be Szegedy's Fourier sigma-algebra, and put

$$U_0 = \mathbb{E}(U \mid \mathcal{F}), \qquad H_0 = \mathbb{E}(H \mid \mathcal{F}).$$

Lemma 4.3 shows that every degree-one coloured density of U, H equals the corresponding density of U_0, H_0. Lemma 4.4 gives one compact group G, one factor map $\pi : A \to G$, and functions $u, h : G \to [0, 1]$ with

$$U_0 = u \circ \pi, \qquad H_0 = h \circ \pi.$$

Since π is a measure-preserving homomorphism, π^d pushes the ultraproduct probability measure on A^d to Haar measure on G^d, and

$$\pi(L_\ell(\mathbf{x})) = L_\ell(\pi(x_1), \ldots, \pi(x_d)).$$

Thus the ultraproduct density equals the compact integral in part (a). By the initial diagonal choice, the ordinary subsequential limit of the finite densities equals the same ultralimit, proving (a). Parts (b) and (c) are exactly the content of Lemma 4.4.

For (d), let $\chi \in \widehat{G}$. Then $\chi \circ \pi$ is a linear character of the ultraproduct. By Szegedy's Proposition 6.1, there are characters $\gamma_n \in \widehat{A_n}$ such that

$$\chi \circ \pi = \lim_{\omega} \gamma_n.$$

Using the Fourier convention stated in the introduction,

$$\begin{aligned} \widehat{h}(\chi) &= \int_G h(y)\overline{\chi(y)}\, dm_G(y) \\ &= \int_A H_0(x)\overline{\chi(\pi(x))}\, dx. \end{aligned}$$

Since $H - H_0$ is orthogonal to $L^2(\mathcal{F})$, and $\chi \circ \pi$ is $\mathcal{F}$-measurable, the last integral equals

$$\int_A H(x)\overline{\chi(\pi(x))}\,dx.$$

The ultraproduct integral of the internal function $H\overline{\chi \circ \pi}$ is

$$\lim_{\omega} \mathbb{E}_{x \in A_n} h_n(x)\overline{\gamma_n(x)} = \lim_{\omega} \widehat{h_n}(\gamma_n).$$

The proof for u_n, u is identical. □

Proposition 4.6 (Weighted compactness in the form used below)**.** *Let $p_n \to \infty$ be primes and $A_n = \mathbb{F}_{p_n}$. Let*

$$u_n = \mathbf{1}_{[1,\lfloor \theta p_n \rfloor]}, \qquad h_n = \mathbf{1}_{F_n},$$

where $0 < \theta < 1$, $F_n \subset A_n$, and

$$|F_n| \le (1-\alpha)p_n.$$

After passing to a subsequence, Proposition 4.5 gives a connected compact abelian group G, Haar measure m_G, a factor map π, and functions $u, h : G \to [0,1]$ such that

$$\int_G u\,dm_G = \theta, \qquad \int_G h\,dm_G \le 1-\alpha,$$

and the coloured counting conclusion of Proposition 4.5 holds for u_n, h_n and u, h.

Proof. Apply Proposition 4.5. Apply it first to the one-form system

$$L_1(x) = x.$$

For the colour u_n, this gives

$$\int_G u\,dm_G = \lim_{n\to\infty} \mathbb{E}_{A_n} u_n = \lim_{n\to\infty} \frac{\lfloor \theta p_n \rfloor}{p_n} = \theta.$$

For the colour h_n, after passing to a further subsequence if necessary,

$$\frac{|F_n|}{p_n} \to \delta$$

for some $0 \le \delta \le 1-\alpha$. Therefore

$$\int_G h\,dm_G = \lim_{n\to\infty} \mathbb{E}_{A_n} h_n = \delta \le 1-\alpha.$$

□

5 The coloured counting statement for the Cayley system

Lemma 5.1 (Hereditary property)**.** *Every subsystem of a Cauchy–Schwarz complexity-one system also has Cauchy–Schwarz complexity at most 1.*

Proof. Fix a form in the subsystem. Take the two classes witnessing complexity at most 1 for that form in the full system and intersect them with the subsystem. Passing to smaller classes can only shrink their rational spans, so the required non-containment conditions remain true. □

Lemma 5.2 (Complexity of the enlarged Cayley system)**.** *For every $K \geq 1$, the system*

$$\mathcal{L}_K = \{x_1, \ldots, x_K\} \cup \{x_i - x_j : 1 \leq i < j \leq K\}$$

has Cauchy–Schwarz complexity at most 1*.*

Proof. Fix $L = x_a$. Put

$$\mathcal{C}_1 = \{x_a - x_j : j > a\} \cup \{x_i - x_a : i < a\},$$

and

$$\mathcal{C}_2 = \mathcal{L}_K \setminus (\{x_a\} \cup \mathcal{C}_1).$$

Every form in $\mathcal{C}_1$ is a difference, so every rational linear combination of forms in $\mathcal{C}_1$ has total coefficient sum zero. The form x_a has total coefficient sum one, hence

$$x_a \notin \operatorname{span}_{\mathbb{Q}}(\mathcal{C}_1).$$

Every form in $\mathcal{C}_2$ has coefficient zero in front of x_a, while x_a has coefficient one in front of x_a. Hence

$$x_a \notin \operatorname{span}_{\mathbb{Q}}(\mathcal{C}_2).$$

Now fix $L = x_a - x_b$, with $a < b$. Let

$$\mathcal{C}_1 = \{M \in \mathcal{L}_K \setminus \{x_a - x_b\} : \text{the coefficient of } x_b \text{ in } M \text{ is } 0\},$$

and

$$\mathcal{C}_2 = \mathcal{L}_K \setminus (\{x_a - x_b\} \cup \mathcal{C}_1).$$

Every rational linear combination of forms in $\mathcal{C}_1$ has coefficient zero in front of x_b, whereas $x_a - x_b$ has coefficient -1 in front of x_b. Thus

$$x_a - x_b \notin \operatorname{span}_{\mathbb{Q}}(\mathcal{C}_1).$$

The forms in $\mathcal{C}_2$ are exactly those forms, different from $x_a - x_b$, with nonzero x_b-coefficient. None of them involves x_a: the only form in $\mathcal{L}_K$ involving both x_a and x_b is $x_a - x_b$, which has been removed. Hence every rational linear combination of forms in $\mathcal{C}_2$ has coefficient zero in front of x_a, whereas $x_a - x_b$ has coefficient one in front of x_a. Therefore

$$x_a - x_b \notin \operatorname{span}_{\mathbb{Q}}(\mathcal{C}_2).$$

This proves the claim. □

Lemma 5.3 (Coloured counting for the local Cayley average)**.** *Let u_n, h_n, A_n, G, u, h be as in Proposition 4.6. Then*

$$\begin{aligned}
&\lim_{n\to\infty} \mathbb{E}_{\mathbf{x}\in A_n^K} \prod_{r=1}^{K} u_n(x_r) \prod_{1\leq i<j\leq K} \big(1 - h_n(x_i - x_j)\big) \\
&= \int_{G^K} \prod_{r=1}^{K} u(y_r) \prod_{1\leq i<j\leq K} \big(1 - h(y_i - y_j)\big)\, dm_G^K(\mathbf{y}).
\end{aligned}$$

Proof. Let $P_K = \{(i,j) : 1 \le i < j \le K\}$. Expand

$$\prod_{(i,j)\in P_K} (1 - h_n(x_i - x_j)) = \sum_{S\subset P_K} (-1)^{|S|} \prod_{(i,j)\in S} h_n(x_i - x_j).$$

For a fixed $S \subset P_K$, the corresponding linear forms are

$$\{x_1, \dots, x_K\} \cup \{x_i - x_j : (i,j) \in S\}.$$

This is a subsystem of $\mathcal{L}_K$, hence has Cauchy–Schwarz complexity at most 1 by Lemmas 5.2 and 5.1. Proposition 4.6 therefore applies to each fixed S, with colour u_n on the coordinate forms and colour h_n on the selected difference forms:

$$\begin{aligned}\lim_{n\to\infty} \mathbb{E}_{\mathbf{x}\in A_n^K} \prod_{r=1}^{K} u_n(x_r) \prod_{(i,j)\in S} h_n(x_i - x_j) \\ = \int_{G^K} \prod_{r=1}^{K} u(y_r) \prod_{(i,j)\in S} h(y_i - y_j)\, dm_G^K(\mathbf{y}).\end{aligned}$$

Summing the finitely many limits over $S \subset P_K$ gives the displayed identity. □

6 Fourier positivity and continuous positive-definite representatives

Lemma 6.1 (Fourier positivity in the compact limit)**.** *Assume, in addition to the hypotheses of Proposition 4.6, that*

$$F_n = -F_n$$

and

$$\widehat{h_n}(\gamma) \ge -\varepsilon_n \qquad (\gamma \ne 0), \qquad \varepsilon_n \to 0.$$

Then the limiting function h satisfies

$$\widehat{h}(\chi) \ge 0 \qquad (\chi \in \widehat{G}).$$

Proof. Let $\chi \in \widehat{G}$. By the spectral compatibility clause of Proposition 4.5, there are characters $\gamma_n \in \widehat{A_n}$ such that

$$\widehat{h}(\chi) = \lim_{\omega} \widehat{h_n}(\gamma_n).$$

For $\gamma = 0$,

$$\widehat{h_n}(0) = \mathbb{E}h_n \ge 0,$$

and for $\gamma \ne 0$ the hypothesis gives

$$\widehat{h_n}(\gamma) \ge -\varepsilon_n.$$

Hence the same lower bound $-\varepsilon_n$ holds for all γ. Taking the ultralimit yields

$$\widehat{h}(\chi) \ge \lim_{\omega}(-\varepsilon_n) = 0.$$

Since $F_n = -F_n$, all finite coefficients $\widehat{h_n}(\gamma)$ are real, and therefore the limiting coefficient is real and nonnegative. □

Lemma 6.2 (Positive Fourier coefficients give a continuous representative)**.** *Let G be a compact abelian group and let $g \in L^\infty(G)$ satisfy*

$$0 \le g \le 1 \qquad \text{and} \qquad \widehat{g}(\chi) \ge 0 \quad (\chi \in \widehat{G}).$$

Then

$$\sum_{\chi \in \widehat{G}} \widehat{g}(\chi) \le 1,$$

where the sum is the supremum over finite subsets of $\widehat{G}$. Consequently

$$\phi(x) = \sum_{\chi \in \widehat{G}} \widehat{g}(\chi)\chi(x)$$

converges absolutely and uniformly, defines a continuous positive-definite function $\phi : G \to [0,1]$, and $\phi = g$ almost everywhere.

Proof. Let $S \subset \widehat{G}$ be finite and let $\eta > 0$. By the Bochner–Fejer kernel theorem for compact abelian groups, applied on the finitely generated subgroup of $\widehat{G}$ generated by S, there is a nonnegative trigonometric polynomial K on G such that

$$K \ge 0, \qquad \int_G K \, dm_G = 1,$$

$$\widehat{K}(\rho) \ge 0 \quad (\rho \in \widehat{G}), \qquad \widehat{K}(\chi^{-1}) \ge 1 - \eta \quad (\chi \in S).$$

See Rudin [12, Chapter 1, Sections 1.4–1.5] for the Bochner–Fejer kernels and their approximation property. Since $0 \le g \le 1$ and $K \ge 0$,

$$0 \le \int_G gK \, dm_G \le 1.$$

Because K is a finite trigonometric polynomial,

$$\int_G gK \, dm_G = \sum_{\rho \in \widehat{G}} \widehat{K}(\rho)\widehat{g}(\rho^{-1}),$$

with only finitely many nonzero summands. All summands are nonnegative. Therefore

$$(1-\eta) \sum_{\chi \in S} \widehat{g}(\chi) \le \int_G gK \, dm_G \le 1.$$

Letting $\eta \downarrow 0$ gives

$$\sum_{\chi \in S} \widehat{g}(\chi) \le 1.$$

Taking the supremum over finite S proves the asserted ℓ^1 bound.

The Fourier series defining ϕ is then absolutely and uniformly convergent, so ϕ is continuous. For arbitrary $x_1, \dots, x_m \in G$ and $c_1, \dots, c_m \in \mathbb{C}$,

$$\sum_{i,j=1}^{m} c_i \overline{c_j} \phi(x_i - x_j) = \sum_{\chi \in \widehat{G}} \widehat{g}(\chi) \left| \sum_{i=1}^{m} c_i \chi(x_i) \right|^2 \ge 0.$$

Thus ϕ is positive-definite. Moreover

$$\widehat{\phi}(\chi) = \widehat{g}(\chi) \qquad (\chi \in \widehat{G}).$$

All Fourier coefficients of $(g-\phi)dm_G$ vanish. Since trigonometric polynomials are uniformly dense in $C(G)$, the measure $(g-\phi)dm_G$ is zero, and $g = \phi$ almost everywhere. Finally, m_G has full support; continuity of ϕ and the almost everywhere bounds $0 \le g \le 1$ imply $0 \le \phi \le 1$ everywhere. □

Lemma 6.3 (The one-set of a Fourier-positive function)**.** *Let $g : G \to [0,1]$ be continuous and positive-definite, and suppose*

$$g(x) = \sum_{\chi \in \widehat{G}} a_\chi \chi(x), \qquad a_\chi \ge 0, \qquad \sum_\chi a_\chi \le 1,$$

with uniform convergence. Put

$$H_g = \{x \in G : g(x) = 1\}, \qquad P = \{\chi \in \widehat{G} : a_\chi > 0\}.$$

If $H_g \neq \varnothing$, then

$$H_g = \bigcap_{\chi \in P} \ker \chi.$$

If $H_g = \varnothing$, then the same conclusion is replaced by the explicit alternative $g(0) < 1$. In particular, whenever H_g is nonempty, it is a closed subgroup of G. If additionally $g \not\equiv 1$, then H_g is a proper closed subgroup.

Proof. If $H_g = \varnothing$, then in particular $g(0) < 1$, since $g(0) = 1$ would put $0 \in H_g$. Now assume $H_g \neq \varnothing$, and choose $x \in H_g$. Then

$$1 = |g(x)| \le \sum_\chi a_\chi = g(0) \le 1.$$

Equality holds throughout. Equality in the triangle inequality implies that every character with $a_\chi > 0$ has the same phase at x; since $g(x) = 1$ is real positive and $\sum a_\chi = 1$, that phase is 1. Thus $\chi(x) = 1$ for every $\chi \in P$, and

$$H_g \subset \bigcap_{\chi \in P} \ker \chi.$$

Conversely, if $y \in \bigcap_{\chi \in P} \ker \chi$, then

$$g(y) = \sum_\chi a_\chi = g(0) = 1,$$

so $y \in H_g$. This proves the equality and shows that H_g is a closed subgroup when nonempty. If $H_g = G$, then $g \equiv 1$. Hence, if $g \not\equiv 1$, a nonempty H_g is proper. □

Lemma 6.4 (Proper closed subgroups of connected compact groups)**.** *If G is a connected compact abelian group and $H < G$ is a proper closed subgroup, then $m_G(H) = 0$.*

Proof. A closed subgroup of a compact group has positive Haar measure only if it is open: this follows, for example, from the Steinhaus theorem, since positive measure gives that $HH^{-1} = H$ contains a neighbourhood of the identity. If H were open, then the quotient G/H would be a finite discrete compact group. The continuous image of a connected group is connected, so G/H would be trivial, forcing $H = G$. Thus a proper closed subgroup cannot have positive Haar measure. □

Lemma 6.5 (Changing a compact representative on a null set)**.** *Let $u : G \to [0,1]$ be measurable with $\theta = \int u\, dm_G > 0$, and let $d\nu = u\, dm_G/\theta$. If two Borel functions $h, h' : G \to [0,1]$ agree Haar-almost everywhere, then for every K,*

$$\prod_{i<j}(1 - h(y_i - y_j)) = \prod_{i<j}(1 - h'(y_i - y_j))$$

for ν^K-almost every $(y_1, \dots, y_K)$.

Proof. Let $E = \{x : h(x) \neq h'(x)\}$, so $m_G(E) = 0$. Since $\nu \ll m_G$, every translate of E is ν-null. For fixed all variables except y_i, the set of y_i such that $y_i - y_j \in E$ is a translate of E, hence ν-null. Fubini gives

$$\nu^K\{\mathbf{y} : y_i - y_j \in E\} = 0$$

for each pair (i, j). A finite union over pairs proves the claim. □

7 Weighted positivity of the limiting integral

Lemma 7.1 (Weighted positivity)**.** *Let G be a connected compact abelian group. Let $u : G \to [0,1]$ be measurable and suppose*

$$\theta := \int_G u\, dm_G > 0.$$

Set

$$d\nu = \frac{u\, dm_G}{\theta}.$$

Let $h : G \to [0,1]$ be continuous and positive-definite, and suppose that

$$\int_G h\, dm_G \leq 1 - \alpha$$

for some $\alpha > 0$. Then, for every $K \geq 1$,

$$\int_{G^K} \prod_{1 \leq i < j \leq K} \big(1 - h(y_i - y_j)\big)\, d\nu^K(\mathbf{y}) > 0.$$

Proof. For $K = 1$ the integral is 1. Assume $K \geq 2$. By Lemma 6.2, applied to h,

$$h(x) = \sum_{\chi \in \widehat{G}} a_\chi \chi(x), \qquad a_\chi \geq 0, \qquad \sum_\chi a_\chi \leq 1,$$

with uniform convergence. Let

$$H_h = \{x \in G : h(x) = 1\}.$$

By Lemma 6.3, either $H_h = \varnothing$, or H_h is a closed subgroup. Since $\int h\, dm_G \leq 1 - \alpha$, $h \not\equiv 1$, so in the nonempty case H_h is a proper closed subgroup. By Lemma 6.4,

$$m_G(H_h) = 0$$

whenever $H_h \neq \varnothing$. The same null conclusion is automatic if $H_h = \varnothing$.

Since $\nu \ll m_G$, every translate of H_h is ν-null. Choose $y_1, \dots, y_K \in \operatorname{supp} \nu$ inductively so that

$$y_i - y_j \notin H_h \qquad (i \neq j).$$

At step r, avoid the finite union

$$\bigcup_{i<r} (y_i + H_h),$$

which has ν-measure zero; because $\nu(\operatorname{supp} \nu) = 1$, this is possible. For the chosen tuple,

$$h(y_i - y_j) < 1 \qquad (i < j).$$

By continuity of h, there are neighbourhoods $V_1, \dots, V_K$ of $y_1, \dots, y_K$, and a number $\eta > 0$, such that

$$1 - h(z_i - z_j) \geq \eta \qquad (z_i \in V_i,\ z_j \in V_j,\ i < j).$$

Each V_i has positive ν-measure because $y_i \in \operatorname{supp} \nu$. Therefore

$$\int_{G^K} \prod_{i<j} (1 - h(y_i - y_j))\, d\nu^K \geq \eta^{\binom{K}{2}} \prod_{i=1}^{K} \nu(V_i) > 0.$$

□

8 Completion of the proof of the one-sided lemma

The contradiction argument begun in Section 3 is now completed.

Assume Lemma 1.2 fails. Then there are $p_n \to \infty$, $F_n \subset \mathbb{F}_{p_n}$, $I_n = [1, \lfloor \theta p_n \rfloor]$,

$$u_n = \mathbf{1}_{I_n}, \qquad h_n = \mathbf{1}_{F_n},$$

and

$$\mathbb{E}_{x_1, \dots, x_K \in I_n} \prod_{i<j} \big(1 - h_n(x_i - x_j)\big) = 0$$

for all n.

By Proposition 4.6, after passing to a subsequence there are G, u, h with

$$\int_G u\, dm_G = \theta, \qquad \int_G h\, dm_G \leq 1 - \alpha,$$

and G connected. By Lemmas 6.1 and 6.2, h may be replaced by a continuous positive-definite representative with $0 \leq h \leq 1$. By Lemma 6.5, this replacement does not change the weighted integrals below.

Rewrite the local average as a global weighted average:

$$\begin{aligned} &\mathbb{E}_{x_1, \dots, x_K \in I_n} \prod_{i<j} \big(1 - h_n(x_i - x_j)\big) \\ &\quad = \left(\frac{p_n}{|I_n|}\right)^K \mathbb{E}_{\mathbf{x} \in \mathbb{F}_{p_n}^K} \prod_{r=1}^{K} u_n(x_r) \prod_{i<j} \big(1 - h_n(x_i - x_j)\big). \end{aligned}$$

By Lemma 5.3,

$$\begin{aligned} &\lim_{n \to \infty} \mathbb{E}_{\mathbf{x} \in \mathbb{F}_{p_n}^K} \prod_{r=1}^{K} u_n(x_r) \prod_{i<j} \big(1 - h_n(x_i - x_j)\big) \\ &\quad = \int_{G^K} \prod_{r=1}^{K} u(y_r) \prod_{i<j} \big(1 - h(y_i - y_j)\big)\, dm_G^K. \end{aligned}$$

Since

$$\frac{|I_n|}{p_n} \to \theta,$$

one obtains

$$\begin{aligned}
&\lim_{n\to\infty} \mathbb{E}_{x_1,\dots,x_K\in I_n} \prod_{i<j}\big(1-h_n(x_i-x_j)\big)\\
&= \theta^{-K}\int_{G^K}\prod_{r=1}^{K} u(y_r)\prod_{i<j}\big(1-h(y_i-y_j)\big)\,dm_G^K\\
&= \int_{G^K}\prod_{i<j}\big(1-h(y_i-y_j)\big)\,d\nu^K,
\end{aligned}$$

where

$$d\nu = \frac{u\,dm_G}{\theta}.$$

By Lemma 7.1, the last integral is strictly positive. This contradicts the assumption that all local averages are zero. Lemma 1.2 follows.

9 Completion of Erdős Problem 42

Theorem 1.1 was reduced to Lemma 1.2 in Section 2, and that lemma has now been proved. Hence, for every $M \geq 1$, choosing

$$K = R(M), \qquad \alpha = \frac{1}{2}, \qquad \theta = \frac{1}{8}$$

and then N sufficiently large gives the desired Sidon set $B \subset [1, N]$ of size M with

$$(A - A) \cap (B - B) = \{0\}.$$

This proves Theorem 1.1.

A Every sufficiently large finite set of integers contains a Sidon subset of size M

Lemma A.1. *For every $M \geq 1$ there is an integer $R(M)$ such that every finite set $X \subset \mathbb{Z}$ with*

$$|X| \geq R(M)$$

contains a Sidon subset of size M.

Proof. A greedy proof is given. Suppose $B \subset X$ is already Sidon and

$$|B| = t.$$

It remains to estimate how many $y \in X \setminus B$ cannot be added to B while preserving the Sidon property.

The set $B \cup \{y\}$ fails to be Sidon only if two sums involving elements of $B \cup \{y\}$ coincide and at least one of the two sums uses y. Since B is Sidon, no equality among old sums is possible. There are two types of forbidden equalities.

First, for $b, b_1, b_2 \in B$, one may have

$$y + b = b_1 + b_2.$$

For each ordered triple $(b, b_1, b_2) \in B^3$, this determines at most one value

$$y = b_1 + b_2 - b.$$

Thus this type forbids at most t^3 values of y.

Second, for $b_1, b_2 \in B$, one may have

$$2y = b_1 + b_2.$$

For each ordered pair $(b_1, b_2) \in B^2$, this determines at most one integer y. Thus this type forbids at most t^2 values.

The remaining possible equality

$$y + b = 2y$$

forces $y = b$, and these values are already excluded by $y \notin B$. The equality $y + b_1 = y + b_2$ forces $b_1 = b_2$ and introduces no forbidden value. Therefore, at stage t, at most

$$t + t^2 + t^3$$

elements of X are unavailable, including the already chosen elements. If

$$|X| > \sum_{t=0}^{M-1} (t + t^2 + t^3),$$

the greedy construction can be continued until it has chosen M elements. Thus one may take

$$R(M) = 1 + \sum_{t=0}^{M-1} (t + t^2 + t^3).$$

□

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